

The Axiom of Choice¹

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Axiom of Choice *If the product of a family of sets is empty then one of the members of the family is empty.*

This axiom is known to be independent of the other axioms of set theory.

Theorem *The following statements are logically equivalent:*

- (a) *the axiom of choice;*
- (b) *surjective functions have right inverse functions;*
- (c) *(strong Zorn's Lemma) an ordered set in which each chain has an upper bound has a maximal element;*
- (d) *(Zorn's Lemma) an ordered set in which each chain has a least upper bound has a maximal element;*
- (e) *(Hausdorff maximal principle) every ordered set has a maximal chain;*
- (f) *(Zermelo well-ordering principle) every set admits a well order.*

Proof *(c) implies (d)* Obvious.

(d) implies (e) Let X be any ordered set. Let C be the set of chains in X (a chain in X is a subset which is linearly ordered by the order of X). Take C to be ordered by inclusion. We must show that C has a maximal element. We would like to apply (d) to C . Take any chain $\mathcal{D}$ in C . Let E be the union of all the sets in $\mathcal{D}$.

I claim E is a chain. For, suppose $x, y \in E$. Then x is in some D_1 in $\mathcal{D}$ and y is in some D_2 in $\mathcal{D}$. Since $\mathcal{D}$ is a chain, either D_1 is a subset of D_2 or D_2 is a subset of D_1 . In either case, we have a D in $\mathcal{D}$ with $x, y \in D$. But D is a chain in X , so either $x \leq y$ or $y \leq x$. So E is a chain.

Clearly then E is a least upper bound for $\mathcal{D}$. So, by (d), C has a maximal element.

(e) implies (f) Let X be any set. Take $\mathcal{W}$ to be the set of pairs $(A, \leq)$ where A is a subset of X and $\leq$ is a well ordering of A . Order $\mathcal{W}$ by $(A, \leq) \leq (A', \leq')$ if and only if A is a subset of A' , the order $\leq$ on A is induced by the order $\leq'$ on A' , and A is an initial segment² of $(A', \leq')$. An initial segment of an initial segment is an initial segment; so $\leq$ is an order on $\mathcal{W}$. By (e), $\mathcal{W}$ has a maximal chain C . Let $Y = \bigcup \{A : (A, \leq) \in C\}$. Define an order on Y by $x \leq y$ iff $x \leq y$ in some A for $(A, \leq) \in C$; using the fact that C is a chain, we see that this is a well-defined order on Y .

I claim Y is well ordered. Take any non-empty subset Z of Y . Then there exists $z \in Z$, so $z \in A$ for some $(A, \leq) \in C$. By definition of the order on Y , we have $\{x \in Y : x \leq z\} = \{x \in A : x \leq z\}$. Now A is well ordered, so $Z \cap \{x \in A : x \leq z\}$ has a least element. So $Z \cap \{x \in Y : x \leq z\}$ has a least element. So Z has a least element. So Y is well ordered.

¹ These notes are essentially those I took at the University of Sydney when Max Kelly lectured to the 1965 Mathematics Honours Year. He learned this treatment at Cambridge.

² This means $x \in A$, $y \in A'$, $y \leq' x$ imply $y \in A$.

It is easy to see that $(A, \leq) \leq (Y, \leq)$ in $\mathcal{W}$ for all $(A, \leq) \in C$. So C together with $(Y, \leq)$ is also a chain in $\mathcal{W}$. But C was a maximal chain in $\mathcal{W}$, so $(Y, \leq) \in C$.

We complete the proof of this part by showing that $Y = X$. For, if not, there is some $x \in X$ not in Y . Let $Y^+ = Y \cup \{x\}$ with order $\leq^+$ extending that of Y by making x the last element. Then $(Y^+, \leq^+)$ is a well order with $(Y, \leq) \leq (Y^+, \leq^+)$. So C together with $(Y^+, \leq^+)$ is a chain in $\mathcal{W}$. By maximality of C , we have $(Y^+, \leq^+) \in C$ contrary to the fact that $(Y, \leq)$ is the last element of C .

(f) implies (a) Take a family of non-empty sets $A_i, i \in I$, with I non-empty. Well order the union U of the sets in the family using (f). Let a_i be the least element of the non-empty subset A_i of U . Then (a_i) is an element of the product.

(a) implies (b) Suppose $f : A \longrightarrow I$ is a surjective function. Then, for $i \in I$, the fibre $A_i = \{a \in A : f(a) = i\}$ is non-empty. So, by the axiom of choice, the product of the sets A_i has an element (a_i) . Define a function $s : I \longrightarrow A$ by $s(i) = a_i$. Then $f \circ s = 1_I$. So s is a right inverse for f .

(b) implies (c) This part needs a lemma.

Lemma Let X be any ordered set and let $\text{Ch}(X)$ be the set of chains in X . There exists no function $f : \text{Ch}(X) \longrightarrow X$ such that $f(A) > a$ for all elements a of all chains A of X .

Proof Suppose we have such a function $f : \text{Ch}(X) \longrightarrow X$. For each $a \in A \in \text{Ch}(X)$, put $A_a = \{x \in A : x < a\}$. For the purposes of this proof, we define a chain A to be *special* when it is well ordered by the ordering of X and $f(A_a) = a$ for all $a \in A$.

I claim that, if A, B are special chains with $B \subseteq A$, then B is an initial segment of A . To see this we may suppose $B \neq A$ since the case $B = A$ is clear. Since A is well ordered, there is a first element a of A which is not in B . Then $A_a \subseteq B$. To prove the claim we shall see that $A_a = B$. Suppose not: since B is well ordered, there is a first element $b \in B$ not in A_a . Then $B_b \subseteq A_a$. But we shall show that $A_a \subseteq B_b$. For certainly $A_a \subseteq B$; yet $b \notin A_a$ and $b \in A$ and A is a chain, so $b \geq a$. So any $x \in A_a$ has $x < a \leq b$; so $x \in B_b$. This proves $A_a = B_b$. Since A, B are special, $a = f(A_a) = f(B_b) = b$. So $a \in B$, a contradiction.

I next claim that, if A, B are special chains, then either $A \subseteq B$ or $B \subseteq A$. Suppose $A \subseteq B$ is false. Then there is a first element $a \in A$ not in B . We shall see that $B = A_a$. Certainly $A_a \subseteq B$. An initial segment of a special chain is a special chain, so A_a is a special chain. By the last paragraph, A_a is an initial segment of B . So either $A_a = B$ as desired or $A_a \subseteq B_b$ for some $b \in B$. In the latter case, $a = f(A_a) = f(B_b) = b$. So $a \in B$, a contradiction. So $B = A_a \subseteq A$.

Now let M be the union of all the special chains in X . I claim M is a special chain. It is well ordered since it is the union of well ordered sets such that, for any two of the sets, one is an initial segment of the other. Also, if $a \in M$ then a is in some special chain A ; so $M_a = A_a$; so $a = f(A_a) = f(M_a)$.

So M is the largest special chain. But $M \cup \{f(M)\}$ is a larger one. This contradiction concludes the proof of the Lemma.

We can now conclude the proof of the Theorem by proving the strong form of Zorn's Lemma given that surjective functions have right inverses. Suppose X is an ordered set such that every chain in X has an upper bound. Assume that X has no maximal element. Let $Y = \{ (A, x) : x \in X, A \text{ is a chain in } X, \text{ and } a < x \text{ for all } a \in A \}$. Define

$$\begin{aligned} h : Y &\longrightarrow \text{Ch}(X) \text{ by } h(A, x) = A, \text{ and} \\ p : Y &\longrightarrow X \text{ by } p(A, x) = x. \end{aligned}$$

I claim h is surjective. Take a chain A . If A has a greatest element a , since a cannot be maximal in X , there exists $x > a$; so $x > b$ for all $b \in A$. If A has no greatest element, let x be an upper bound for A ; so $x > a$ for all $a \in A$. In each case we have $(A, x) \in Y$ and $h(A, x) = A$.

By (b), the function h has a right inverse $k : \text{Ch}(X) \longrightarrow Y$. Put $f = p \circ k$ which means $k(A) = (A, f(A))$ for all chains A . So $f : \text{Ch}(X) \longrightarrow X$ is a function with $f(A) > a$ for all $a \in A$. This is a contradiction to the Lemma.

Quod Erat Demonstrandum.

Appendix of definitions

An *order* on a set P is a relation $\leq$ satisfying the following conditions

reflexive $x \leq x$

transitive $x \leq y$ and $y \leq z$ imply $x \leq z$

antisymmetric $x \leq y$ and $y \leq x$ imply $x = y$.

An *ordered set* is a set P together with a chosen order. Each subset of an ordered set is ordered by the restriction of the order relation to the subset.

An order is called *linear* (or *total*) when, for all x, y , either $x \leq y$ or $y \leq x$. A *chain* in an ordered set P is a linearly ordered subset of P .

An *upper bound* for a subset S of an ordered set P is an element u of P such that $s \leq u$ for all $s \in S$. A *least upper bound* (or *supremum*) for a subset S of P is an upper bound u such that, if v is an upper bound for S then $u \leq v$. Similarly, *greatest lower bound* (or *infimum*) is defined by reversing order.

An element m of an ordered set P is *maximal* when, for all $x \in P$, if $m \leq x$ then $m = x$. Similarly, *minimal* element is defined by reversing order.

A *smallest* (or *first* or *bottom*) element of an ordered set P is an element z such that $z \leq x$ for all $x \in P$. Similarly, *largest* (or *last* or *top*) element is defined by reversing order. Largest elements are maximal, but not every maximal element is largest in general. Largest elements are unique (if one exists), whereas an ordered set can have many maximal elements.

A *well order* on a set P is an order such that every non-empty subset has a smallest element.

The *product* of a family $(X_i : i \in I)$ of sets X_i indexed by the elements i of a set I is the set X whose elements are families (x_i) of elements $x_i \in X_i$.